The Natural Geometry Hierarchy: A Synthesis of the GeoVac Quantum-Chemistry Arc*

J. Loutey[1]

[1] *Independent Researcher, Kent, Washington*

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The Geometric Vacuum (GeoVac) framework discretizes non-relativistic electronic structure on *natural geometries*: for each separable quantum system there is a coordinate system in which the Coulomb singularities become boundary conditions and the Laplace–Beltrami operator's quantum-number structure (labels, degeneracies, selection rules) is carried exactly by a sparse lattice whose nodes are those labels. This synthesis assembles the framework's quantum-chemistry arc—the twelve Group 2 papers spanning one-electron atoms, the two-center one-electron ion, the two-electron atom, the two-center two-electron molecule, the core–valence and polyatomic extensions, the genuine two-center tensor census, the transcendence frontier at the third center, and the isoenergetic Sturmian secular equation—into a single coherent narrative organized by the *natural geometry hierarchy*. The hierarchy is a grid indexed by the number of nuclear centers and the number of electrons: S^3 (Fock) for atoms [21], the prolate spheroid for H_2^+ [23], hyperspherical coordinates for He [25], molecule-frame hyperspherical coordinates for H_2 [26], and composed fiber bundles for core–valence and polyatomic systems [27, 29]. Three threads run through the arc. *First*, an algebraic-first program: the angular content is computed exactly from Gaunt coefficients and Wigner $3j$ symbols at every level, the prolate-spheroidal electron–electron repulsion is replaced by an algebraic Neumann expansion [24], the radial solvers reduce to three-term Laguerre recurrences, and the split-region Legendre nuclear coupling terminates exactly via the $3j$ triangle inequality [26]; what currently survives as quadrature (Z_{eff} screening, the adiabatic potential curves, the hyperradial coupling) is named explicitly, and finding algebraic replacements is an ongoing goal rather than a deficiency. *Second*, two structural negative theorems delimit the design space: in the single-n D-matrix *approximation* to a single-center molecular Sturmian basis with a shared momentum scale, the one-electron Hamiltonian and overlap obey the two-term identity $H_{ij} = -\frac{1}{2}p_0^2 S_{ij} + (1-\beta_j)V_{ij}$ and share a single orthogonal $SO(4)$ Wigner-D congruence, so the generalized eigenvalues are independent of the internuclear distance and cannot bind unless the Sturmian scale $\beta_k(R)$ itself becomes R-dependent [22]. (Paper 8's own scope remark is explicit that this delimits that approximation and is *not* a no-go for Sturmian molecular binding: the genuine cross-n Coulomb–Sturmian overlap breaks the shared congruence and supplies the R-dependence algebraically—see the Paper 60 entry below, where the method binds.) And a graph built by concatenating atom-centered lattices has an R-independent intra-atomic kinetic energy, producing a monotonically attractive potential-energy surface with no equilibrium [35]. A later genuine two-center census sharpens where the sparsity lives: evaluated over orbital products that span two nuclei the Gaunt l-selection rule fails and only the axial m-rule survives, so the angular sparsity is an *atomic-sector* property—the abelian residue of the molecular point group [30]. *Third*, honest accuracy ceilings: the framework reaches machine precision for H_2^+, 0.022% (properly variational, raw $l_{\max} = 7$) down to 0.004% (a non-variational cusp extrapolation) for He, 99.97% of D_e (0.053 mHa) for H_2 via an explicit-r_{12} extension of the prolate spheroidal engine with exact algebraic integrals (99.81% / 0.32 mHa via re-conditioned CI at the variational optimum $\alpha = 1.40$)—and 96.0% (cusp-corrected) in the molecule-frame hyperspherical treatment—two routes to the same system, not a matched pair, and H_2's natural geometry is the prolate spheroid—and—in the composed architecture, whose accuracy bottleneck is the Phillips–Kleinman pseudopotential—5.3%, 11.7%, and 19.4% equilibrium-bond-length error for LiH, BeH_2, and H_2O respectively. The classical-solver investigation is complete: its accuracy ceilings are characterized, and the framework is positioned as a zero-parameter, structurally sparse *research instrument*—a computationally principled alternative whose primary downstream value (qubit-Hamiltonian sparsity for quantum simulation) lives in the quantum-computing arc—not as a replacement for production quantum chemistry. The discrete graph and the continuous Laplace–Beltrami operator are dual descriptions connected by proven conformal maps; the interpretive question of which is more fundamental is left open.

I. INTRODUCTION

The Group 3 foundations arc of GeoVac establishes that the hydrogen atom's momentum-space Schrödinger equation maps, by Fock's 1935 stereographic projection [1], onto the free Laplacian on the unit three-sphere

* Group 2 synthesis in the Geometric Vacuum series

S^3, and that the quantum-number labels (n, l, m) of a discrete graph are exactly the labels of the S^3 harmonics, on which the S^3 Laplace–Beltrami operator acts as the diagonal $SO(4)$ Casimir at every cutoff [19–21] (the graph's edge Laplacian does not converge to that operator; corrected 2026-09-25). That arc answers a question about a single one-electron atom. The present synthesis takes up the question the foundations arc opens but does not close: what is the right discretization for a *molecule*, or for an atom with *more than one electron*?

The organizing answer is the *natural geometry principle*, stated first in Paper 11 [23]: every separable quantum system has a natural lattice, namely the label set and channel structure of the coordinates in which the problem separates (for S^3 this is the label lattice, not a discretization of the Laplace–Beltrami operator, whose graph Laplacian does not converge to it, Paper 7). For a one-center one-electron system that coordinate system is S^3; for a two-center one-electron system it is the prolate spheroid; for a one-center two-electron system it is the hyperspherical coordinate system; for a two-center two-electron system it is the molecule-frame hyperspherical system; and for systems with electrons on separated length scales (a tight core and a diffuse valence shell) the natural object is a *composed* geometry—a fiber bundle in which each electron group inhabits its own natural coordinate system, coupled by screening functions and pseudopotentials. Crucially, this is not a linear tower in which each level subsumes the previous one: the Level-4 two-electron coordinates (R_e, α) are simply undefined for a single electron, so the hierarchy is a two-dimensional grid indexed by (number of centers) × (number of electrons) [27].

This grid structure is the reason the chemistry arc is not a single algorithm but a sequence of natural geometries, each with its own separation of variables, its own algebraic content, and its own accuracy ceiling. Table I summarizes the hierarchy and the headline result at each level; the body of this synthesis unpacks each row, the algebraic structure that makes it work, and the two proven theorems that mark the boundaries of what natural geometry can and cannot do.

GeoVac is presented throughout as a discretization framework that exploits natural geometry, not as a new foundation for quantum mechanics. The discrete graph and the continuous Laplace–Beltrami operator are dual descriptions connected by the proven conformal maps of the foundations arc, and the interpretive question of ontological priority is left to the reader. The framework's concrete advantages—a zero-parameter construction from nuclear charges and geometry alone, angular-momentum selection rules baked into the basis, exact algebraic matrix elements, and structural sparsity—are independent of that interpretive question and are the properties this synthesis foregrounds.

II. LEVEL 1–2: FROM THE ATOM TO THE DIATOMIC ION

A. The atomic baseline and the bond sphere

The one-electron atomic baseline is the S^3 graph of the foundations arc: on the discrete (n, l, m) label lattice $-\Delta_{S^3}$ is the diagonal $SO(4)$ Casimir $n^2 - 1$, so through Fock's energy shell the Rydberg spectrum holds at every cutoff; the graph Laplacian itself does not converge to the S^3 operator, and its extremal eigenvalue matches the ground state only by construction through κ [20, 21]. The natural first question for chemistry is whether the same S^3 object handles a *molecule*. Paper 8 [22] pursues this directly: it shows that a diatomic molecule is geometrically a single S^3 with two weighted poles—the *bond sphere*—in which the internuclear distance R encodes as a polar angle γ through $\cos\gamma = (p_0^2 - p_R^2)/(p_0^2 + p_R^2)$, and all cross-atom matrix elements are $SO(4)$ Wigner D-matrix elements evaluated at the bond rotation $e^{i\gamma A_y}$. The geometry is consistent, and it yields exact selection rules for σ-bonds: the bond rotation preserves $m = 0$ within $l = 1$ and does not mix s and p states, $D^2_{10,10}(\gamma) = 1$ and $D^2_{00,10}(\gamma) = 0$ for all γ [22].

B. The Sturmian structural theorem (a guardrail)

The bond sphere construction also produces the arc's first proven *negative* theorem, and it is load-bearing for everything that follows. In the molecular Sturmian basis [10] with a shared momentum scale p_0, the one-electron Hamiltonian matrix is

$$H_{ij} = -\frac{p_0^2}{2} S_{ij} + (1 - \beta_j) V_{ij}, \tag{1}$$

where S_{ij} is the overlap matrix, V_{ij} the nuclear-attraction matrix, and β_j is the Sturmian eigenvalue of orbital j. The Hamiltonian is *not* a scalar multiple of S: both (H, S) carry their entire R-dependence through the same $SO(4)$ Wigner-D congruence $D^{(n)}(\gamma(R))$, so the generalized eigenvalues $\varepsilon_k = \frac{1}{2}p_0^2 - p_0^2/\beta_k$ (which equals $-p_0^2/2$ only in the degenerate case $\beta_k = 1$) are independent of the bond angle γ, hence independent of the internuclear distance R [22]. A basis that produces R-independent eigenvalues cannot describe binding, which is by definition an R-dependent lowering of the energy—so *within this approximation* binding is excluded. Paper 8 sharpens this further: for a heteronuclear diatomic with charge ratio $Z_A/Z_B > 1$ there is no atom-centered shared-p_0 Sturmian basis that binds both centers simultaneously, and the dual-scale repair drives the two p_0 values toward a geometric mean that eliminates the scale separation [22].

This theorem is a guardrail in the project's sense: it rules out an entire class of single-center, shared-exponent molecular encodings *built on the single-n D-matrix approximation* (nested hyperspherical, unified-basis, single-

TABLE I. The GeoVac natural geometry hierarchy for quantum chemistry. Each level is the coordinate system in which the corresponding system separates; the result column reports the headline figure at its stated tier, with the source paper. Composed (Level 5) results are limited by the Phillips–Kleinman pseudopotential, the characterized accuracy ceiling of the composed architecture (Sec. V).

Level	System (centers, electrons)	Natural geometry	Headline result
1	atom (1, 1)	S^3 (Fock projection)	$< 0.1\%$ (one-electron)
2	H_2^+ (2, 1)	prolate spheroid	machine precision (spectral Laguerre)
3	He (1, 2)	hyperspherical	0.022% (2D var. raw) to 0.004% (cusp-extrapolated, non-variational);
4	H_2 (2, 2)	molecule-frame hyperspherical	96.0% of D_e ($l_{\max} = 6$, 61 channels, cusp-corrected)
4*	H_2 (2, 2)	prolate spheroidal	99.97% of D_e (0.053 mHa) via explicit-r_{12} CI (exact integrals); 99.81%
4N	LiH (2, 4)	full mol-frame ($SO(12)$)	$R_{\rm eq} \approx 63.5\%$ ($l_{\max} = 2$, 2D variational; unbound D_e, no PK)
5	LiH (core+valence)	composed (L3 $\oplus$ L4)	$R_{\rm eq}$ 5.3% (l-dependent PK)
5	BeH_2 (polyatomic)	composed + 1-RDM exchange	$R_{\rm eq}$ 11.7%
5	H_2O (triatomic)	composed (five-block, uncoupled)	$R_{\rm eq}$ 19.4%

p_0 Sturmian CI), and one resolution—reintroduce the R-dependent spectrum by moving to the prolate spheroidal coordinates where the diatomic actually separates—is the natural-geometry principle in action.

a. Scope correction (Paper 8 `rem:overlap_imposed`*).* The theorem's backing test *imposes* $S_{AB} = f\, D^{(n)}$ by construction, so what it verifies is a property of the bond-sphere D-matrix model, not of the genuine cross-n Coulomb–Sturmian basis (where, e.g., $\langle \chi^A_{1s} | \chi^B_{2s} \rangle = -0.47$ exceeds the within-n diagonal $+0.35$ at $R = 3.015$). A cross-n overlap breaks the shared congruence, so binding is *not* excluded for the genuine basis: Paper 8's `cor:binding` records that the R-dependence is then supplied *algebraically* by the Shibuya–Wulfman overlap (a function of $s = p_0 R$), needing no continuous two-center PDE, and Paper 60 exhibits the binding directly. Prolate spheroidal geometry remains the route this arc took; it is not the only one the theorem permits.

C. The prolate spheroidal lattice for H_2^+

Paper 11 [23] implements that resolution. The two-center one-electron problem separates in prolate spheroidal coordinates (ξ, η, ϕ) [3], and the separated equations discretize on a prolate spheroidal lattice with a spectral angular solver (Legendre basis) and a spectral radial solver [18], with Brent root-finding in the separation parameter $c^2 = -R^2 E_{\rm elec}/2$ to enforce self-consistency. The method has zero free parameters: all inputs are nuclear charges, the internuclear distance, and the basis resolution. With the spectral Laguerre radial solver at $n_{\rm basis} = 20$, the H_2^+ ground-state energy is $E_{\min} = -0.602634214494936$ Ha against the reference -0.6026342144949 Ha [2]—agreement to 3.6×10^{-14} Ha, i.e. to the precision at which that reference is quoted, making it the most accurate single result in the chemistry arc—with the correct dissociation to H+p as $R \to \infty$ [23]. The spectral solver achieves a 250× dimension reduction and a 270× speedup, and reaches machine precision, against a finite-difference baseline whose 1.01% error is shown to be a discretization artifact, not an intrinsic limit of the basis [23].

The algebraic structure here is the first concrete instance of the arc's central theme. For σ states ($m = 0$), the spectral Laguerre matrix elements are computed entirely from three-term recurrence relations on Laguerre moment matrices, eliminating numerical quadrature from the radial solver completely (the overlap is exact, $S_{mn} = \delta_{mn}/2\alpha$, and the kinetic and potential matrices are finite sums of moments, reproduced to $\sim 10^{-12}$ against quadrature) [23]. For π and δ states ($m \neq 0$) an associated Laguerre basis absorbs the centrifugal $1/x$ singularity via a partial-fraction decomposition and a Stieltjes integral recurrence, reducing the non-algebraic content of the entire solver to a *single* transcendental seed, the exponential-integral value $e^a E_1(a)$; all higher moments follow from the rational recurrence $S_0^{(k)}(a) = \Gamma(k) - a\, S_0^{(k-1)}(a)$ for $k \geq 1$ [23]. This is the exchange-constant taxonomy of the foundations arc made concrete: the irreducible transcendental content of the diatomic radial problem is one seed in the *composition* tier, tagged to a specific integral, rather than diffuse numerical error [28].

The separation constant itself is an exact label. Paper 11 shows that in any product (Laguerre × Legendre) basis the matrices of H and of the Erikson–Hill separation-constant operator Λ are Atkinson operator determinants. They therefore commute at every truncation, measured to 10^{-14}. This is the two-centre counterpart of the $SO(4)$ Casimir being exact on the S^3 labels. The commutation and the determinant theorem are prior art; the finite-truncation reading was not found in the literature. Λ interpolates from L^2 at the united atom to the Runge–Lenz (parabolic) label at the separated atoms, as checked quantitatively. Two limits are stated with it. As a two-electron configuration selector, Λ beats orbital-energy ordering 8–10× for H_2, but only in one basis at one geometry, and natural orbitals beat it by about 10×. On the Route C LiH orbital set (product orbitals, but not tensor-closed) the exactness is lost and the selector fails [23]. Tensor closure alone would not rescue it: a one-body probe finds that the short-range penetration of the

Li core is what breaks the label for the σ virtuals [23, 29].

III. LEVEL 3: THE TWO-ELECTRON ATOM AND THE CUSP

A. Algebraic V_{ee} on the prolate spheroidal lattice

Before leaving the diatomic geometry, Paper 12 [24] adds the second electron and confronts the electron–electron repulsion $\langle\phi_i|1/r_{12}|\phi_j\rangle$ over the six-dimensional two-particle configuration space. Standard numerical quadrature on a grid converges slowly because of the Coulomb singularity at $r_{12} = 0$ and saturates near 80% of the exact H_2 dissociation energy regardless of basis size. The algebraic replacement is the Neumann expansion [4] of $1/r_{12}$ in prolate spheroidal coordinates, which decomposes each six-dimensional integral into a finite sum of products of one-dimensional auxiliary integrals computable by recurrence relations: the angular moments $C_l(q)$, the exponential moments $A_n(\alpha)$, and the first- and second-kind Legendre moments satisfy exact three-term recurrences, so the V_{ee} matrix is exact within the Neumann truncation order with no six-dimensional quadrature and no fitted parameters [24]. That property is established in the σ sector; the azimuthal extension described below does not inherit it. One seed resists: the log-singular second-kind moment B_l is evaluated by a single one-dimensional adaptive quadrature, its recurrence being numerically unstable at the base case. A selection rule $C_l(q) = 0$ for $q < l$ or $(q - l)$ odd truncates the effective sum automatically.

On H_2 at $R = 1.4011$ bohr the algebraic Neumann V_{ee} reaches 92.4% of the exact D_e with 72 basis functions, against 80.1% for numerical V_{ee} at the same basis size—a uniform 12–20 percentage-point improvement at every basis size from 1 to 72 functions, and the Neumann result plateaus, confirming that the V_{ee} *integration* error has been eliminated [24]. The remaining 7.6% gap is a basis-space limit rather than an integration error—and, as Paper 12 now establishes, a *configuration*-space one. That basis is φ-independent, so it spans only $m_1 = m_2 = 0$; but a $^1\Sigma_g^+$ state constrains only the total $M = m_1 + m_2$, so the π^2 and δ^2 configurations—which carry the angular correlation—are absent by construction. Restoring them at $|m| \leq 1$ in the same basis, with the same Neumann kernel, reaches 99.1% of D_e in the original monomial basis—a gain of 11.6 mHa against the 0.34 mHa available from near-tripling the σ basis. Re-conditioning that same span (Laguerre × Legendre, high-precision assembly) then climbs to 99.81% (0.32 mHa, inside chemical accuracy) at the variational optimum $\alpha = 1.40$, and an explicit-r_{12} extension of the same engine—every r_{12} matrix element reduced to the same algebraic Neumann kernels, no quadrature—reaches 99.97% (0.053 mHa) [24]. The $|m| = 2$ sector was numerically unstable in the differentiation-based radial engine, so the δ contribution ($\approx$0.5 mHa) was first fixed by an independent Gaussian-basis route; the quoted figure carries a stability envelope, the basis being strongly linearly dependent at that size. Paper 12 has since generalised the same A_l, B_l, X_l recurrences to the associated Legendre functions, giving a moment-recurrence engine that is stable in the δ sector—where the differentiation route drove the energy far below the variational floor—and reproduces the sound σ and $|m| \leq 1$ results; the $\mu > 0$ channels are therefore now recurrence-based (recurrence-stable and—since the B-seeds were closed in $\{E_1, \gamma, \ln\}$—fully quadrature-free).

Withdrawn. Earlier versions of this synthesis reported Paper 12's original diagnosis—that the gap was the electron–electron cusp demanding non-analytic $r_{12}^{1/2}$ and $r_{12} \ln r_{12}$ terms, and that this identified hyperspherical coordinates as the next natural geometry. Paper 12 has withdrawn both, and the correction reached further than this section: Paper 18's Level-4 subsection [28] had built an *irreducibility* argument on the same saturation—that the product basis is "structurally blind to three-body coalescence"—and that argument has been withdrawn with it, taking the qualitative Level-3/Level-4 distinction with it. What survives is narrower and genuinely independent: the cusp keeps its place as the framework's embedding-tier transcendental in Paper 18's taxonomy, which is a statement about the cusp in any product basis and never rested on the H_2 residual. It is simply not what capped this calculation.

B. The hyperspherical lattice for helium

Paper 13 [25] introduces hyperspherical coordinates for helium—on their own grounds, the atomic coalescence being genuinely three-body, rather than as a consequence of the H_2 residual above. In hyperspherical coordinates [8, 9] the He problem becomes a coupled-channel expansion in which the angular eigenvalue problem depends parametrically on the hyperradius R, so the channel functions vary with R; this coupled-channel structure has the form of a fiber bundle and is the mathematical prototype for electron–nuclear coupling in molecules. The electron–electron cusp at $\alpha = \pi/4$, $\theta_{12} = 0$ becomes a boundary condition rather than a singularity [25].

The accuracy results here are reported at carefully separated tiers, and the synthesis preserves the distinction. The original single-channel adiabatic solver gives $E = -2.9052$ Ha, nominally 0.05% from the exact -2.9037 Ha, but it is *non-variational*: it lies below the exact energy through a fortuitous cancellation of adiabatic and finite-difference error, and the paper labels it as such [25]. The properly variational results come from a two-dimensional variational solver that treats the hyperradius and hyperangle simultaneously, avoiding the adiabatic approximation: 0.022% raw at $l_{\max} = 7$ (a genuine variational upper bound) and 0.004% with a Schwartz cusp correction at $l_{\max} = 4$ (a non-variational extrapolation that lies marginally below the exact energy) [25]. A

separate, fully zero-parameter route is configuration interaction in the fixed $k = Z$ hydrogenic basis on the S^3 labels, with the exact diagonal one-body operator and exact rational Slater integrals: it reaches 1.88% above exact at $n_{\max} = 7$, a floor set by the missing continuum of the bound-state basis rather than by the cusp [25]. (An earlier "graph-native" 0.216% added a non-physical $+\frac{1}{16}$ [retracted 2026-09-25: he-graph-native-hop-headline] one-body graph hop to helium's Hamiltonian and is withdrawn as a helium result, as are the readings built on it—an "over-binding" H^- and a "graph-validity boundary" $Z_c \approx 1.84$.) Each route exercises a different [retracted 2026-09-25: he-graph-native-hop-headline] part of the algebraic registry: the angular sector is exact throughout (Gaunt integrals $G(l, k, l') = 2\,(l\,k\,l'\,|\,0\,0\,0)^2$, $SO(6)$ Casimir eigenvalues $\nu(\nu + 4)/2$, closed-form perturbation coefficients), while the adiabatic potential curves $U_{\text{eff}}(R)$ and the parametric angular eigenvalues $\mu(R)$ remain point-by-point diagonalizations—numerical content named explicitly as a current, not fundamental, feature [25, 28].

Paper 13 also documents an instructive negative result: in the *adiabatic* approximation, increasing $l_{\max}$ beyond 0 worsens the energy, because the finite-difference treatment of the boundary singularities $l(l + 1)/\cos^2\alpha$ and $l(l+1)/\sin^2\alpha$ degrades faster than the partial-wave content improves; the adiabatic channel truncation has an intrinsic floor near 0.19%–0.20%. The 2D variational solver is what breaks that floor [25]. The same fiber-bundle structure, coupled to a nuclear lattice, delivers an end-to-end demonstration: ab initio rovibrational constants for H_2 with zero spectroscopic input, the fundamental $\nu_{01} = 4157$ cm^{-1} (0.1%) and the rotational constant $B_e = 59.5$ cm^{-1} (2.2%) [25].

IV. LEVEL 4: THE TWO-CENTER TWO-ELECTRON MOLECULE

The two-center two-electron molecule H_2 [5] combines all five Coulomb singularities and had resisted a single natural coordinate system. Paper 15 [26] introduces molecule-frame hyperspherical coordinates $(R_e, \alpha, \theta_1, \theta_2, \Phi)$ parameterized by the dimensionless ratio $\rho = R/(2R_e)$, where R is the internuclear distance and R_e the electronic hyperradius. The electron–electron cusp appears at $\alpha = \pi/4$, $\theta_{12} = 0$ *independent of molecular geometry*—a structural difference from prolate spheroidal coordinates, where the same locus is not a coordinate surface—a difference in where the cusp sits, which does not translate into an accuracy advantage—and the angular eigenvalue problem depends on (R, R_e) only through ρ, collapsing a two-dimensional parameter sweep to one dimension [26].

A coupled-channel expansion in body-frame partial waves (l_1, m_1, l_2, m_2) with the gerade constraint $l_1 + l_2$ even and $m_1 + m_2 = 0$, solved with the same two-dimensional (R_e, α) variational solver that avoids the adiabatic approximation, recovers 96.0% (cusp-corrected; ∼95% pure-variational) of the exact D_e [6] at $l_{\max} = 6$ with σ and π channels (61 channels; a complete-basis extrapolation reaches ∼97%) [26].

Withdrawn. Earlier versions read this as exceeding the 92.4% of prolate spheroidal CI (Paper 12) and confirming a cusp-resolution advantage for the hyperspherical geometry. The comparison was not like-for-like: the 96.0% includes π channels and the 92.4% does not. At matched $m_{\max}$ no comparison is available: Paper 12's own basis with $|m| \leq 1$ reaches 99.1% in its original monomial basis and 99.81% re-conditioned, but those figures and the 96.0% differ in truncation, cusp treatment and solver class alike. Neither paper claims a coordinate-system advantage over the other, and the two published figures are not a matched pair in any case: the 96.0% is $l_{\max} = 6$ with a Schwartz cusp correction and a different solver class. What both papers do agree on is the physics that matters here: the $m \neq 0$ channels carry the angular correlation, and neither geometry gets it without them. The framework extends to heteronuclear diatomics: for HeH^+ a charge-center origin shift and removal of the gerade constraint recover 93.1% of the exact D_e at $l_{\max} = 4$ with $\sigma + \pi$ channels in the *adiabatic* treatment [26]; this is a non-variational over-binding figure—the consistent two-dimensional variational treatment recovers only ∼5% of D_e, so this is not parity with the H_2 result.

The algebraic content reaches its cleanest statement here. The nuclear potential coupling is a split-region Legendre expansion whose sum over Gaunt integrals terminates *exactly* at $k = l' + l$ by the $3j$ triangle inequality $|l' - l| \leq k \leq l' + l$—no truncation parameter, exact to machine precision in all regimes [26]. This is the arc's prime example of a continuous integration replaced by a finite algebraic sum once the right structure is identified. What remains numerical is named: the Z_{eff} screening correction (Gauss–Legendre quadrature), and the angular eigenvalue sweep at each ρ, which is a point-by-point diagonalization because the split-region structure is piecewise smooth [26, 28].

Paper 15 is careful about variational hygiene. The adiabatic approximation systematically *over*-estimates D_e by roughly 11% relative to the 2D variational solver, and at $l_{\max} = 4$ with $m_{\max} = 1$ the adiabatic D_e exceeds the exact value, violating the variational principle; the 2D result is the correct bound, and the ∼ 10 percentage-point gap between them is the kinetic-energy cost of non-adiabatic coupling [26]. The convergence also shows an even–odd staircase: even-$l_{\max}$ increments add large gains (the gerade constraint forces direct nuclear coupling only at even $l_{\max}$) while odd increments are negligible—a selection-rule signature, not a numerical accident [26].

V. LEVEL 5: COMPOSED GEOMETRIES AND THE ACCURACY CEILING

A. The composition principle

Systems with core electrons resist a single natural geometry because core and valence electrons occupy different length scales, and—as noted in Sec. I—the hierarchy does not collapse downward (Level-4 coordinates are undefined for one electron) [27]. Paper 17 [27] introduces *composed* natural geometries: fiber bundles in which each electron group inhabits its own natural coordinate system, coupled by screening functions and ab initio Phillips–Kleinman (PK) pseudopotentials [17] derived entirely from the core wavefunction. For LiH, the Li^+ $1s^2$ core is solved by the Level-3 hyperspherical method, yielding a screening function $Z_{\text{eff}}(r)$ and the PK parameters; the two valence electrons are solved by the Level-4 molecule-frame method with Z_{eff} injected; and the potential-energy surface is scanned by a ρ-collapse adiabatic method. The construction uses *zero* experimental molecular data—all PK inputs are atomic (the core wavefunction, the ionization energy, the nuclear charges) [27].

B. Results and the PK accuracy ceiling

The composed LiH potential-energy surface gives $R_{\text{eq}} = 3.21$ bohr at $l_{\max} = 2$ (6.4% error versus the experimental 3.015 bohr [16], improved from 17% in the LCAO baseline) and $\omega_e = 1471$ cm^{-1} (4.6%); an l-dependent PK variant, which restricts the projector to the $l = 0$ core channels, reduces the bond-length error to 5.3% [27]. The same parameter-free formula transfers to BeH_2 and H_2O: with full one-reduced-density-matrix inter-fiber exchange, BeH_2 reaches 11.7% bond-length error (matching a single-parameter fitted exchange model at zero free parameters), and the uncoupled H_2O construction reaches 19.4% (improved from 26% by general solver evolution—*not* by the PK-consistency fix, which the uncoupled PES does not invoke) with a correct qualitative surface [27].

Two findings establish that the PK pseudopotential is the *characterized accuracy ceiling* of the composed architecture, not a tunable knob. First, the PK barrier is essential, not cosmetic: removing it collapses R_{eq} to 0.88 bohr, so PK is what creates the bonding-versus-Pauli-repulsion balance that produces an equilibrium at all [27]. Second, the residual error is structural. Across all four solver×PK combinations the bond-length error saturates near 5–6% at the $l_{\max} = 2$ operating point, and the $l_{\max}$ drift (the systematic outward creep of R_{eq} with partial-wave order) is not removed by the 2D variational solver, by the l-dependent PK, or by an R-dependent PK whose reference length cannot be derived from atomic quantities without circular reasoning [27]. The paper attributes this to a deep correlation wall—labeled W1e, the composed architecture's instance of the framework's multi-focal composition wall pattern—and documents that single-mechanism repairs each saturate well below closure: a bonding-orbital PK extension at a 43% ceiling, explicit-core Hartree pressure at $\sim 26\%$, basis enlargement at $\sim$10%, with Schmidt orthogonalization and core correlation each contributing negligibly [27]. This wall is later relocated from the correlation treatment to the Hamiltonian specification itself, and given a group-theoretic mechanism, in Sec. VI.

C. Equilibrium without PK: the Level-4N control

That PK is a *cost of factorization* rather than a missing physics term is established by the Level-4N control, the exact N-electron generalization in which all four LiH electrons share one S^{11} Hilbert space under $SO(12)$ with S_4 antisymmetry [27]. At $l_{\max} = 2$ (12 S_4 channels, angular dimension 2592) the Level-4N surface has a genuine equilibrium near $R_{\text{eq}} \approx 1.0$–$1.1$ bohr ($\approx 64\%$ error) with *no* PK—demonstrating that binding exists without the pseudopotential, and that the $l_{\max} = 2$ $S_4[2,2]$ angular basis is genuinely insufficient for LiH binding rather than the encoding being wrong. The composed geometry trades this exact inter-group antisymmetry for a 144× angular compression (cost scaling $l_{\max}^2$ versus $l_{\max}^{11}$), which is what makes the composed architecture the production operating point despite the 5–6% ceiling [27]. The reason the trade is forced and not a matter of convenience is a structural obstruction: exact antisymmetry requires a shared coordinate system, while composition factorizes the wavefunction into incompatible coordinate systems—so the three tested PK alternatives (node exclusion, a fiber-bundle Berry connection, spectral orthogonality) each fail for the same reason [27].

VI. THE PK-FREE ALTERNATIVE AND THE FCI ENCODINGS

A. Balanced coupled composition (Paper 19)

Paper 19 [29] pursues the natural follow-on: can the PK pseudopotential be replaced by explicit cross-block physics? The balanced coupled Hamiltonian adds cross-center nuclear attraction V_{ne} via a multipole expansion with exact Gaunt termination (the multipole sum terminates at $L_{\max} = 2l_{\max}$ by the same $3j$ selection rules, with no truncation error), restoring the cross-block one-body coupling that the composed architecture lacks; these missing one-body terms are the Shibuya–Wulfman integrals of Coulomb Sturmian theory [11, 29]. The result is the PK-free alternative for single-point energies at fixed geometry: LiH four-electron FCI reaches 0.20% energy error at $n_{\max} = 3$ (down from 1.8% at $n_{\max} = 2$), using analytical incomplete-gamma cross-center integrals at machine precision [29]. The structural cost is geometric rather than energetic: the equilibrium bond length still drifts to $R_{\text{eq}} = 3.28$ bohr ($\approx 8.8\%$ error), though the drift

per $n_{\max}$ step is about three times smaller than the PK $l_{\max}$ drift [29]. The honest reading is that the energy axis converges through $n_{\max} = 3$ while the geometry drifts—but not that the energy axis is simply the good one: at $n_{\max} = 4$ the model crosses *past* exact (over-binding), so the fixed-geometry error is not monotonically improvable either [29]. The balanced coupled Hamiltonian is therefore suited to single-point quantum simulation at fixed geometries rather than to full potential-energy surfaces, with that non-monotonicity disclosed. The drift is not a physics wall. An all-electron prolate-spheroidal (Level 2) treatment gives $R_{\text{eq}} = +0.2\%$, and its analytic, resolution-of-identity–free explicit-r_{12} energy, -8.04905 Ha, is the corpus's best LiH energy. It is variational and cross-checked by Monte-Carlo. That route buys energy per qubit, not sparsity. Its qubit Hamiltonian is dense within M_L, with 8,370 Pauli terms at $Q = 20$, ten times the composed LiH count at $Q = 30$ [29]. Paper 19 also records a clean negative: cross-block electron–electron integrals *without* the matching cross-center V_{ne} are energetically unbalanced (29% error, worse than PK), and a harmonic-driven orthogonalization wall blocks second-row hydrides such as NaH, which PK cross-center coupling reduces but does not close [29]. That NaH wall is crossed in Paper 58 (next subsection), by evaluating the integrals over genuine two-center orbital products rather than atomic-sector surrogates—at the cost of the label-generated sparsity [30].

B. The abelian residue: why the sparsity is atomic-sector (Paper 58)

Paper 58 [30] closes the question the balanced-coupled work leaves open—whether the label-generated sparsity and the genuine cross-center physics can coexist—and the answer is group-theoretic rather than technical. Every prior molecular builder evaluates its integrals in the *atomic sector*: each orbital product entering an integral sits on a single nucleus, and inter-nuclear coupling enters through a surrogate (the composed builder's Z_{eff} screening; the balanced builder's displaced-nucleus one-body term, with both orbitals still on one center). Because neither ever forms a genuine two-center orbital product, the Gaunt l-selection rule is exact for them *by construction*, and the sparsity follows.

Evaluated instead over orbital products that genuinely span two nuclei, $\langle\chi^A| \cdot |\chi^B\rangle$, in exact rational arithmetic (Mulliken auxiliary functions for the overlaps; a hydrogenic eigen-trick for the cross-center one-body term, certified entry-wise by an exact bra/ket identity), the census measures that the Gaunt l-selection rule does *not* survive while the axial m-rule does, and the symmetry-permitted electron-repulsion density inflates 13.8× over the builder at $n_{\max} = 2$. An internal theorem supplies the mechanism. The label m names the abelian symmetry both centers share—rotation about the internuclear axis is a single global one-parameter group—while l names irreducible representations of a rotation group attached to one center. The intersection of the two centers' rotation groups is exactly the axial $U(1)$, whose character group is $\mathbb{Z}$ by Pontryagin duality; abelian selection rules are additive charge conservation and survive, whereas non-abelian Clebsch–Gordan selection needs the full sphere and does not. *The surviving cross-center sparsity is the abelian residue of the molecular point group, readable in advance.*

Two re-readings follow. First, the sparsity previously charged to Löwdin orthogonalization (a 17.9× Pauli inflation) was largely a bill already owed: the genuine tensor's own permitted density inflates by the same order ($\sim$15×) before any transformation, so orthogonalization is not, in the bulk, the cause of the density blamed on it. Second, four earlier relocations of the non-orthogonality each moved the cost without reducing it—the cost measures the physical asymmetry of a two-center problem—and equivalent-atom-swap tapering *reduces* the qubit saving while inflating the Pauli count, because identifying blocks by permutation destroys the per-block abelian labels that generate the sparsity. The sparsity-versus-binding trade is a conservation law, not a defeat.

The corresponding gain is a panel-verified demonstration. All-electron twelve-electron NaH—the framework's hardest documented binding failure, which six Phillips–Kleinman modifications, an explicit-core Hartree–Fock treatment, and a DMRG cross-check had each left unbound—binds at $R_{\text{eq}} = 3.736\,a_0$ (+4.8% versus experiment), recovering 91.1% of the in-basis FCI binding from three fragment-native determinants, and with *fewer* correlation degrees of freedom than the runs that failed—which is what carries the inference. This localizes W1e to the Hamiltonian specification rather than the correlation treatment: the wall was the atomic-sector construction, not the determinant expansion. The demonstration deliberately forfeits the sparsity, because the sparsity and the atomic-sector evaluation are the same choice.

A subsequently supplied two-center electron-repulsion engine closes all four integral classes in exact form (at transcendence weight one, π-free over $\{E_1, \ln, \gamma\}$), which upgrades the $(AA|BB)$ block of the census from *counted* to *decided*: all 195 symmetry-permitted entries are proven genuinely nonzero by Lindemann–Weierstrass separation, with no accidental zeros and none missed by the counting. For the three genuinely cross classes—79.5% of the tensor—Paper 58 states a three-rung position that is *materially better than counted*: the seed functions $\{1, e^{-\lambda R}, E_1(\mu R), \ln R\}$ are linearly independent over $\mathbb{Q}(R)$ by Liouville–Rosenlicht, so every permitted entry is *functionally* nonvanishing (unconditional) and generically nonzero (measured); but it is pointwise *decidable* only conditional on a Schanuel-type independence hypothesis for the E_1 and γ values. The block is therefore blocked from DECIDED by open transcendence, *not* by missing labour—E_1-value independence at algebraic points is an open problem and γ is not known even to be irrational [30]. Weight one is not the same as de-

cidable.

C. The transcendence frontier at the third center (Paper 59)

Paper 58 closes the two-center tensor in closed form; Paper 59 asks what happens at the third. Read in momentum space, the three-center electron-repulsion integral is a two-scale Bessel moment whose two Fock scales define a quartic curve: the two-center case is genus zero (elementary, weight-one $\{E_1, \ln, \gamma\}$), and the third center raises the genus to one. Its period is a complete elliptic integral, it satisfies an explicit fourth-order Picard–Fuchs equation, and the connection is *irreducible* (a Fourier–Laplace transform of an irreducible rank-one connection), so no closed form arises through factorization [31]. The collinear observable is certified to 66 digits and guarded integer-relation searches are decisively negative through weight three at clean heights (the companion Paper 61 [32]); the finite closed form remains open, and the paper says so.

Two consequences for this arc. The polyatomic question below is thereby sharpened from "can the genuine-integral route be extended?" to a statement about transcendence class: the extension is not blocked by missing algebra but by a genus jump, and any closed-form polyatomic tensor is all-or-nothing (a part-closed-form, part-fitted tensor is not decidable at all). And the same obstruction gates explicitly-correlated (F12) evaluation, because the difficulty of a two-electron integral lives in the densities, not in the operator between them.

That density-not-operator reading is borne out constructively at two centers, where the genus is zero: Paper 12 carries out the explicit-r_{12} (James–Coolidge) CI of a two-center *four-electron* reference *without* a resolution of identity. The genuinely four-body disjoint-pair bridge $f_{12} f_{34}\, r_{13}^{-1}$ and the three-body triangle $f_{ik} f_{jk}\, r_{ij}^{-1}$—the two integral classes that force an RI in Gaussian F_{12} theory—both reduce by the *same* Neumann machinery, screened-Coulomb (Yukawa) kernels included, which is exactly the operator-independence at work; the full analytic $2 \times 2 \to E_{R12}$ is then assembled and validated against variational Monte-Carlo for both an $e^{-\gamma r}$ and a cusp-correct $r\, e^{-\gamma r}$ geminal ($E_{R12} = -7.942$ and -7.917 Ha, variational) [24]. This is a proof of concept (an ionic single-ζ reference, not a spectroscopic LiH): the deliverable is the RI-free reduction, not the correlation energy recovered. The gate is therefore specifically the three-center genus jump, not explicit correlation as such.

D. The isoenergetic secular equation as a quantum algorithm (Paper 60)

Paper 60 carries the Sturmian thread into the quantum-algorithm setting and, in doing so, exhibits the binding the scoped guardrail above does not forbid. Naively encoding shared-scale Coulomb–Sturmians in L^2 inflates the block-encoding 1-norm (the non-orthogonal overlap must be Löwdin-inverted, and $S^{-1/2}$ is dense); Avery's *isoenergetic* (potential-weighted) reformulation removes the metric *for atoms* entirely—a metric-free standard eigenproblem whose eigenvalues are the scaling parameters $p_\kappa = \sqrt{-2E}$, from which the energies follow directly [33]. Its 1-norm grows more slowly than the matrix dimension over every basis the paper can compute ($\|M\|_1 \sim K^{0.82}$ fitted on $K = 74$–164, helium), and *increasingly* so: the local slope falls monotonically to 0.766 by $K = 514$, so no window fit is stable and the object is closer to $K^p \log K$ than to a power law. The nuclear diagonal $T^0 = Z \sum_\nu R_\nu$ is the slowest-growing block and the only one derived rather than measured — $Z\sqrt{2K}\ln(K/2)$, asymptotic exponent $1/2$ — while neither grouping of the pure-number block T' is superlinear ($K^{0.88}$ full, $K^{0.94}$ off-diagonal). *The substantive finding is that the advantage belongs to the basis-growth rule, not to the construction*: it holds at fixed $l_{\max}$ and weakens under full-shell growth, where the pure-number block reaches $K^{1.07}$ while the total exponent rises only to 0.911 — still sublinear. The cheap rule carries an accuracy floor bracketed at $[6.47, 6.62]$ mHa — the bracket the paper quotes, since its fit family approaches the floor from below and a single value would be a lower estimate rather than a central one. *The mechanism of that floor is now pinned, and it is neither angular truncation nor the span*: the metric-free form exists only when the basis scale is locked to the eigenvalue, $\lambda = p_\kappa = \sqrt{-2E}$ — an algebraic identity, since $E = -\lambda^2/2$ is exactly what cancels the L^2 metric — and that lock is not the variational optimum. Solving the *identical* span variationally with the scale freed reaches 1.28 mHa at $K = 130$ against 7.46 mHa locked, and 0.15 mHa of the independently known s-limit at $K = 136$; the price is the whole encoding advantage, $\|\cdot\|_1$ from $K^{0.72}$ to $K^{2.75}$ on the s-only comparison ladder ($K = 21$–136, not the headline window). Metric-free posing and accuracy floor are one fact rather than two, both being the scale lock; the sublinearity is not part of that identity, since it belongs to the basis-growth rule. The floor is moreover a *ground-state* pathology, and Avery's own mechanism explains why: a Goscinskian $1s^2$ configuration pins both electrons to one exponent, where an excited configuration gets two free from $n_a \neq n_b$. Since $\|M\|_1$ does not depend on which root is extracted, every ^1S state costs the same to encode — and at $K = 452$ — the largest basis where both roots were computed, full $s{+}p{+}d{+}f$ — the ground state sits $4.28\times$ above chemical accuracy while $2\,^1S$ sits $1.08\times$; and on the *s-only* ladder at $K = 105$ the posing cost falls by $4.3\times$, $3.0\times$ and $2.6\times$ across the first three rungs. (The two are different ladders and the paper says so; quoting one against the other is the defect this sentence was rewritten to remove.) (Superseded 2026-09-07: the earlier $K^{0.84}$, "drifts upward", T^0-at-$K^{0.70}$ and T'-superlinear-at-$K^{1.05}$ [retracted 2026-09-08: p60-tprime-superlinear] readings were measured on a truncated radial domain whose error grew with the fit

variable.) That split is what decides the molecular case. For molecules the Shibuya–Wulfman metric returns, and the paper measures rather than assumes its cost: the conditioning grows polynomially, at a band-limited concentration rate that is the Kac–Murdock–Szegő asymptotic (what is the paper's own is the *identification* of the metric as such a finite section, not the law). Of its four levers, the gerade sector's basis-independent conditioning is an *equivalent-center* property, and a symmetry-unique heavy atom reinstates the raw *growth* there; but a band-Toeplitz preconditioner removes the conditioning cost without requiring equivalent centers, reaching water's A_1 block when the band is aligned to the symbol's null direction (in the unrotated frame the same band is worse than no treatment at all), and a direct block-encoding of the preconditioned metric takes the penalty from n^3 to n. Established on s-sector shared-scale bases at $M = 2$ and non-collinear $M = 3$; the collinear case is open. What remains standing is the *locality* cost, capped by the symbol's other pole, and the ℓ-block sparsity loss, which is structural rather than a conditioning effect. The N-electron 1-norm question is answered in the negative: the atomic sublinearity is a property of the isoenergetic secular matrix's single-center diagonal, not of the basis, and it does not survive two centers.

For the guardrail above this is the operative datum: the method *binds*—H_2^+ at 1.3% s-only underbinding and a minimal-basis H_2 at $E_{\mathrm{tot}} \approx -1.09$ Ha, in a common-scale basis—because the genuine overlap supplies the R-dependence the single-n approximation lacked. What it does not do is recover GeoVac's angular sparsity, which is the trade Paper 58 states as a conservation law.

E. Graph-native FCI for atoms

The atomic FCI paper [34] solves multi-electron atoms directly on the sparse S^3 graph Hamiltonian using Slater–Condon rules [13, 14] with exact R^k radial integrals and Gaunt coefficients, with Slater determinants indexed as sorted tuples of spin-orbital labels (relational lattice indexing). Helium with its physical Hamiltonian reaches 1.93% at $n_{\max} = 5$ and 1.88% at $n_{\max} = 7$ with exact rational Slater integrals, above the Hartree–Fock limit because the fixed-exponent bound-state basis omits the continuum (the earlier 0.26% and 0.216% headlines used a hybrid one-body operator with a non-physical graph hop and are withdrawn as helium results); lithium 1.03% (at $n_{\max} = 5$) and beryllium 0.71% (at $n_{\max} = 4$, with 487,635 determinants) [34]. The Hamiltonian is sparse (fill fraction 0.68% for He at $n_{\max} = 5$, 0.12% for Li at $n_{\max} = 4$) with a sub-quadratic nonzero-count growth exponent $\alpha \approx 1.4$–1.5. The paper documents an instructive variational subtlety: a hybrid one-body operator that mixes graph Laplacian off-diagonals (carrying $\kappa = -1/16$) lowers the He energy only because those off-diagonals are a hop absent from the atom's Hamiltonian, and violates the variational principle for three-electron lithium, so the exact hydrogenic one-body operator is required for monotonic convergence [34]. The dominant remaining error for heteronuclear LiH at $n_{\max} = 3$ is basis truncation, not integral incompleteness—the basis-set superposition error there exceeds the experimental binding energy, so the raw binding number is reported with its counterpoise correction [15] and its honest caveat [34].

F. The graph-concatenation theorem (a second guardrail)

The molecular FCI paper [35] supplies the arc's second proven negative theorem. A topological LCAO architecture that builds a molecular graph by concatenating atom-centered hydrogenic lattices (via Mulliken cross-nuclear attraction, Slater–Condon repulsion, and overlap bridges) is shown to have an intra-atomic graph-Laplacian kinetic energy that is *completely R-independent* by construction: the atomic lattice structure does not change as the nuclei approach. The balanced Hamiltonian therefore produces a monotonically attractive potential-energy surface with no equilibrium geometry—the binding energy decreases smoothly from 0.464 Ha at $R = 1.5$ bohr to 0.042 Ha at $R = 5.0$ bohr, with a virial ratio far from unity [35]. The paper documents a 29-version diagnostic arc that falsifies basis-set superposition error as the geometric driver, rules out orbital-exponent relaxation, and—citing the Sturmian structural theorem of Paper 8—rules out every single-center Sturmian and bond-sphere alternative *of the single-n D-matrix form* as a repair; a single-parameter Fock-weighted kinetic correction partially restores an equilibrium ($R_{\mathrm{eq}} \approx 3.8$ bohr) but cannot fit R_{eq} and D_e simultaneously [22, 35].

Together the two negative theorems—Sturmian structural (shared $SO(4)$ Wigner-D congruence $\Rightarrow$ R-independent eigenvalues, Paper 8) and graph concatenation (R-independent kinetic energy $\Rightarrow$ no equilibrium, the molecular FCI paper)—are guardrails: they explain *why* the natural-geometry hierarchy is necessary. A molecule must be discretized in coordinates where the separation introduces R-dependence (prolate spheroidal, molecule-frame hyperspherical, or composed), because the two obvious shortcuts (one shared-exponent single-center basis in the single-n D-matrix approximation; one concatenated graph) cannot bind in the forms analysed. The shared-exponent leg is the scoped one: the genuine cross-n basis escapes it algebraically (Paper 8 `cor:binding`; Paper 60), at the cost of the qubit-encoding sparsity that motivated the restriction.

VII. THE ALGEBRAIC-FIRST THREAD

A single discipline runs through every level of the hierarchy: replace continuous integration with algebraic evaluation wherever the structure allows, and name what re-

mains. This is the chemistry-arc instance of the framework's prime directive (the discrete angular/selection-rule structure is exact at all levels and must never be approximated; the radial/parametric content is level-dependent and a legitimate target for numerical improvement). Table II catalogues the state of the algebraic registry across the arc.

The pattern is consistent with the foundations-arc taxonomy of Paper 18 [28]. The angular sector is intrinsic (rational, π-free). The diatomic radial seed $e^a E_1(a)$ and the Slater-integral seeds are composition-tier transcendentals tied to a specific integral. The electron–electron cusp is the embedding-tier exchange constant. And the items still in the numerical column—the Z_{eff} screening, the adiabatic curves, the hyperradial coupling—are precisely the places where an algebraic structure has not yet been found, which the project records as an ongoing goal rather than a deficiency: the Neumann expansion (Paper 12) and the split-region Legendre termination (Paper 15) are existence proofs that quadrature walls in this arc *can* dissolve once the right structure is identified [24, 26].

VIII. POSITIONING: A RESEARCH INSTRUMENT

The classical-solver investigation summarized in this synthesis is complete: the natural-geometry hierarchy has been built level by level, its algebraic content catalogued, and its accuracy ceilings characterized. The framework's value proposition is best stated against those ceilings honestly.

A. What the framework offers

The construction is *zero-parameter*: every result in Table I takes as input only nuclear charges, geometry, and a basis resolution—no fitted potentials, no empirical screening constants, no experimental molecular data (the PK pseudopotential of Paper 17 is derived entirely from the atomic core wavefunction) [23, 27]. Angular-momentum selection rules are baked into the basis, so the matrix elements that vanish by symmetry are never computed, and the surviving ones are exact algebraic quantities. The Hamiltonians are structurally sparse, with sub-quadratic nonzero growth [34]. These structural properties—not interpretive claims about the nature of quantum mechanics—are the framework's actual advantages.

B. Where the headline value lives

The arc's strongest computational payoff is not the classical ground-state energies but the qubit Hamiltonians they imply. The angular sparsity that this synthesis documents at the integral level is the structural reason the composed-geometry qubit encodings achieve favorable Pauli-term scaling for variational quantum simulation; that result, its scaling exponents, and its molecular benchmarks belong to the quantum-computing arc (Group 4) and are referenced here, not re-derived. Paper 58 [30] sharpens the scope of that payoff: the l-selection component of the sparsity is an atomic-sector property, exact for the composed encoding precisely because it evaluates per center, so the favorable scaling characterizes small systems evaluated in the atomic sector rather than genuine multi-center integration. This synthesis establishes the chemistry-side foundation of that payoff: the natural-geometry hierarchy, the algebraic-first integral structure, and the two negative theorems that justify the composed encoding.

C. The benchmarking discipline

Where the framework is compared to other methods, the comparison is to the strongest available baseline, and unfavorable comparisons are stated plainly. The composed potential-energy surfaces carry 5–19.4% bond-length error, well above production quantum-chemistry accuracy; the honest reading is that the framework is a computationally principled *alternative* offering sparsity, scaling, and structural insight, not a replacement for correlated electronic-structure methods. The papers report the proven negatives—the Sturmian and concatenation theorems, the PK ceiling, the W1e wall, the failed PK alternatives and cusp treatments—as a first-class part of the corpus, because the characterization of what natural geometry *cannot* do is as load-bearing as what it can [22, 27, 35].

IX. OPEN QUESTIONS AND OUTLOOK

Three structural questions remain open at the close of the classical-solver arc.

a. The composed accuracy ceiling (W1e). The composed architecture saturates near 5–6% bond-length error at its $l_{\max} = 2$ operating point. Whether a composition rule could close W1e without sacrificing the 144× angular compression—i.e. without reverting to the exact-but-expensive Level-4N encoding—was the central open chemistry question of the classical-solver arc; Paper 58 answers it in the negative and gives the reason [30]. The l-selection sparsity is exact only in the atomic sector—it is the abelian residue that survives genuine two-center evaluation—so closing the bond requires forming the two-center orbital products that destroy it. W1e is thereby relocated from the correlation treatment (a DMRG cross-check reproduces FCI bit-identically without binding) to the Hamiltonian specification itself, and the sparsity-versus-binding trade is a conservation law rather than a missing integral. What remains open is quantitative:

TABLE II. The algebraic-first registry across the chemistry arc. *Algebraic* = closed-form or exact-recurrence from quantum-number labels with no quadrature; *algebraic (one seed)* = exact up to a single tagged transcendental; *numerical* = currently requires quadrature or point-by-point diagonalization, named as a present (not fundamental) feature.

Level / object	Status	Mechanism (paper)
Angular coupling (all levels)	algebraic	Gaunt / Wigner $3j$; selection rules [12, 25, 26]
H_2^+ radial, σ ($m = 0$)	algebraic	three-term Laguerre recurrence [23]
H_2^+ radial, π/δ ($m \neq 0$)	algebraic (one seed)	assoc. Laguerre + seed $e^a E_1(a)$ [23]
Prolate spheroidal V_{ee} (σ)	algebraic (one seed)	Neumann recurrences + 1-D quadrature for B_l [24]
Prolate spheroidal V_{ee} ($\lvert m\rvert \geq 1$)	quadrature	same kernel, ordered-ξ integral by spectral panels [24]
He hyperradial S, K matrices	algebraic	three-term Laguerre recurrence [25]
Nuclear coupling (Level 4)	algebraic	split-region Legendre, $3j$-triangle termination [26]
Cross-center V_{ne} (balanced)	algebraic	multipole, exact Gaunt termination $L_{\max} = 2l_{\max}$ [29]
Slater F^k integrals (graph-native)	algebraic	exact rational S^3 node overlap [25, 34]
Z_{eff} screening	numerical	Gauss–Legendre quadrature [26, 27]
Adiabatic potential $U(R)$, $\mu(R)$	numerical	point-by-point diagonalization [25]
Hyperradial coupling (Levels 3–4)	numerical	FD / spectral radial solve [25, 26]

how much of the residual a genuine-integral construction recovers across a range of systems, of which NaH (+4.8%) is the first data point.

b. Algebraic replacement of the remaining quadrature. The numerical column of Table II—Z_{eff} screening, the adiabatic potential curves, the hyperradial coupling—is the standing target of the algebraic-first program. The existence proofs (Neumann V_{ee}, split-region Legendre termination) show such replacements are achievable in principle; whether the adiabatic curves $\mu(R)$, which Paper 13 records as genuinely transcendental in R, admit an algebraic-implicit form (defined by a polynomial equation over a known coefficient ring) is open.

c. Second-row and polyatomic extension. The balanced coupled Hamiltonian hit an orthogonalization wall for second-row systems such as NaH [29]; the genuine two-center construction of Paper 58 crosses it (NaH binds at +4.8%), at the cost of the sparsity [30]. The polyatomic H_2O construction validates only bond–bond coupling, with bond–lone and lone–lone couplings unphysical at the composed level [27]; extending the genuine-integral route past two centers is the open polyatomic question—now sharpened by Paper 59 from an algebra question to a transcendence one: the third center raises the genus from zero to one, the resulting connection is irreducible, and the finite closed form is open [31]. Both remain named boundaries of the current encoding rather than refutations of the hierarchy.

None of these open questions destabilizes the structural results of the arc. The natural-geometry hierarchy is established level by level; the algebraic-first content is catalogued; the two negative theorems are proven; and the accuracy ceilings are characterized. The open questions are frontier questions about how far the framework's sparsity can be pushed before its accuracy ceiling binds, not foundational concerns.

X. CONCLUSION

The Group 2 quantum-chemistry arc of GeoVac is the systematic construction of the natural-geometry hierarchy for electronic structure. Each level is the coordinate system in which a given system separates: S^3 for atoms [21], the prolate spheroid for H_2^+ (machine precision) [23], hyperspherical coordinates for He (0.022% variational raw / 0.004% non-variational cusp; 1.88% fixed-k hydrogenic CI) [25], molecule-frame hyperspherical coordinates for H_2 (96.0% of D_e, cusp-corrected) [26]—though H_2's natural geometry is the prolate spheroid, where an explicit-r_{12} CI reaches 99.97% (0.053 mHa) [24]—and composed fiber bundles for core–valence and polyatomic systems (5.3%, 11.7%, 19.4% bond-length error for LiH, BeH_2, H_2O) [27]. The arc is held together by an algebraic-first discipline—exact angular coupling, algebraic Neumann V_{ee} [24], Laguerre recurrences, and $3j$-terminating Legendre expansions [26]—and is bounded by two structural negative theorems: a single-center shared-p_0 Sturmian basis cannot bind a molecule *in the single-n D-matrix approximation* [22] (the genuine cross-n basis does bind, algebraically, at the cost of the sparsity—[33]), and a concatenated atom-centered graph has no equilibrium [35]. Those theorems are why the hierarchy is the route this arc took; the first is a scoped guardrail rather than an unconditional no-go.

The framework is a zero-parameter, structurally sparse research instrument whose accuracy ceilings are now characterized and whose strongest downstream payoff—qubit-Hamiltonian sparsity for quantum simulation—lives in the quantum-computing arc. The discrete graph and the continuous Laplace–Beltrami operator are dual descriptions connected by proven conformal maps; the framework's concrete advantages do not depend on which description one regards as fundamental, and that interpretive question is left open.

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